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# Teachers' academic motivation and student procrastination behaviour: mediating effects of emotion regulation and study habits

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## Abstract

**Purpose** The present study aimed to analyze the effectiveness of external and personal regulatory mechanisms in reducing procrastination behavior among university students. For this purpose, the role of teachers' academic motivation is worthwhile in shaping the learning environment and reducing procrastination, with a focus on the mediating roles of emotion regulation and study habits considered imperative.

**Research design/method** By employing a quantitative, cross-sectional research design, data were collected from a sample of 210 teachers working in universities located in Multan-Pakistan via convenient sampling, yielding a usable response rate of 70.28%.

**Data analysis & study results** The gathered data was analyzed using Smart PLS 4 software. Results from the structural equation model (PLS-SEM) indicated that teachers' academic motivation significantly affects student procrastination. However, when mediated by emotion regulation and study habits, the study confirmed the statistically significant mediating effect of both mediators in mitigating procrastination among students.

**Findings** Findings of the present study suggested that enhancing teachers' motivation may be more effective when mediated with emotional regulation and effective study habits. Emotionally regulated students and those with effective study habits were less likely to procrastinate during their academic endeavor. The findings highlighted the need for teacher training programs to develop and promote motivation and support students' emotional and academic regulation to enhance learning outcomes.

**Practical implications and recommendations** Based on the above findings, the study suggested that enhancing teacher motivation is more impactful when paired with strategies to foster student emotion regulation and study habits. It also recommended educational policies prioritizing academic content and personalized student engagement strategies to minimize procrastination and promote academic success. Programs should be implemented with a focus on promoting motivation while enabling teachers to support students' emotional and self-regulatory needs for better academic outcomes.

**Keywords** Teacher motivation, Teacher educational support, Procrastination behavior, Emotion regulation, Study habits, Higher education, Educational strategies

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## Introduction

Education is universally recognized as a cornerstone for societal development, a powerful mechanism to enhance intellectual, social, and economic growth. At the tertiary level, universities are pivotal in nurturing human capital and equipping individuals with the skills required to contribute to national progress [1]. In the context of developing nations like Pakistan, higher education assumes even greater significance, as it directly influences workforce productivity and competitiveness on a global scale. As Pakistan endeavors to align with the Sustainable Development Goals (SDGs) for 2030, its higher education system must address critical challenges, including behavioral barriers that hinder academic success [2]. Despite its centrality to national growth, Pakistan's education sector faces persistent issues that undermine its potential. Among these, "Academic Procrastination" has emerged as a significant challenge, particularly in universities, where it adversely affects student outcomes [3]. Procrastination, a multifaceted behavior characterized by delaying tasks, leads to reduced academic performance, heightened stress, and missed opportunities for skill development [4]. While extensive research has explored the internal, student-driven causes of procrastination, such as lack of self-regulation or poor time management, the external, teacher-driven factors still need to be explored. As facilitators of the academic environment, teachers play a crucial role in mitigating or exacerbating procrastination behaviors through their motivation, teaching strategies, and support mechanisms [5].

Current research on procrastination has predominantly focused on students' psychological traits and study behaviors, neglecting the influence of external factors like teachers' academic motivation. Moreover, existing studies often overlook the interplay between external and internal mechanisms, such as how teachers' motivation interacts with students' emotion regulation and study habits to reduce procrastination. This gap is particularly critical in Pakistan, where high levels of procrastination are reported among university students, coupled with a lack of structured interventions addressing this issue [6]. Addressing this gap is essential for designing comprehensive strategies to improve student engagement and academic success, aligning with Pakistan's educational and developmental goals.

The present study offers a novel contribution to higher education and behavioral research by examining the combined influence of teachers' academic motivation and students' internal regulatory mechanisms such as emotion regulation and study habits on reducing procrastination. Because it shifts the focus from predominantly student-centric causes of procrastination to the underexplored role of teacher-driven factors, particularly

teachers' academic motivation, in shaping student behaviors. It also introduces a novel framework by examining the mediating roles of emotion regulation and study habits, offering insights into how internal regulatory mechanisms can enhance the impact of external influences on mitigating procrastination. Providing region-specific insights fills an essential gap in the literature and offers findings that can inform local and regional educational policies. By adopting a systems perspective, this research not only bridges existing gaps but also provides actionable insights for stakeholders in higher education.

## Literature review

This section reviewed a comprehensive literature on teachers' academic motivation, students' procrastination, emotion regulation and study habits.

### Theoretical background

Procrastination is a common yet complex behavior deeply rooted in human psychology and often leads to significant adverse outcomes. According to Klein [7], the Latin origin of the word procrastination comes from the combination of two words: 'pro' (forward) and 'crastinus' (belonging to tomorrow). Procrastination is typically defined as "a tendency to delay an activity or decision" [8, 9], often in a way that defies logic. This delay may be irrational but pervasive, impacting both personal and academic spheres. Rozental and Carlbring [10] emphasize that procrastination is not simply a delay tactic but an automatic, habitual problem of avoiding important tasks, even when there are clear consequences.

In academic settings, the procrastination phenomenon is well understood through the lens of the "theory of Planned Behavior" proposed by Icek Ajzen [11]. TPB posits that intention drives behavior, influenced by attitudes, subjective norms, and perceived behavioral control. This theory best suits to explore this phenomenon because it emphasizes that teachers' academic motivation can affect students' attitudes toward procrastination, creating positive norms and enhancing their control over academic tasks. Solomon and Rothblum [12] argue that procrastination must be understood through multiple dimensions: behavioral, emotional, and cognitive. Neglecting any of these facets makes it difficult to fully grasp the impact of academic procrastination on students' success. This phenomenon not only hinders individual academic achievement but also undermines broader educational and national goals by negatively affecting students' intellectual development and productivity [13]. Over time, procrastination becomes a self-perpetuating cycle that erodes self-esteem and academic motivation, leading to further delays and under-achievement.

Moreover, procrastination can manifest at different times and in varying frequencies, often becoming a life-long habit if not adequately addressed. Sirois et al., [14]. Svartdal and his colleagues explained in their findings that students caught in a cycle of procrastination experience personal costs, such as stress, lower academic or work performance, and strained relationships due to broken commitments and unfulfilled obligations [15]. Studies consistently show that students who procrastinate are more likely to experience academic failure, lower self-esteem, and reduced engagement with learning [16].

In Pakistan, procrastination is particularly prevalent among university students. Research shows that approximately 61% of students exhibit some level of procrastination, while 20% engage in it regularly [17]. This widespread tendency towards procrastination is compounded by academic challenges such as poorly structured tasks, inadequate guidance, and a lack of time-management skills. Such academic environments can foster procrastination by creating conditions where students feel overwhelmed or unsupported, leading them to delay tasks that seem insurmountable [18].

In contrast, procrastinating students often need more motivation, delay academic tasks without immediate fear of consequences, and develop poor academic habits over time [19]. Steel, [20], over time, this behavior can become habitual, resulting in a cycle of procrastination that can be difficult to break. The pervasive nature of procrastination and its detrimental effects make it a critical concern in higher education. Effective interventions must address both procrastination's emotional and cognitive aspects to break this cycle of delay and underachievement.

### Hypothesis development

#### *Teachers' academic motivation and students' procrastination behavior*

In the pursuit of providing quality education and addressing the psychological factors that lead to procrastination, the role of professional teachers is paramount. Teachers are considered professionals when prepared, actively involved, motivated, and willing to engage in teaching and learning activities [21]. Highly motivated teachers are more successful in carrying out essential duties, such as preparing for exams, organizing projects, and assigning homework. These actions create a structured environment that helps reduce student procrastination. Empirical evidences suggest that when teachers are unmotivated, incompatible academic behaviors, such as students giving up quickly or becoming disengaged, often emerge [22]. Thus, a teacher's motivation for academic activities is critical for making the learning process more efficient and effective, promoting student engagement,

and significantly reducing tendencies toward procrastination [23].

Teacher motivation is a decisive external factor that helps mitigate procrastination among students. In this context, motivation is often extrinsically regulated and can be divided into three subcategories: external regulation, introjected regulation, and identified regulation [24]. External regulation occurs when rewards drive motivation or to avoid punishment. Introjected regulation is more internalized but involves acting to prevent feelings of guilt or anxiety or to gain social approval. Identified regulation occurs when the external rewards align with the individual's values and are personally significant. These forms of extrinsic motivation play a vital role in shaping students' behavior, especially in academic environments where procrastination is a prevalent issue [24].

As Codina et al. [25] suggest, motivation is not just about external rewards or pressure but also an internal drive that compels individuals to take purposeful actions. This psychological drive is crucial for both students and teachers, whether conscious or unconscious. Motivated teachers foster an engaging and supportive classroom environment, promoting student motivation, reducing procrastination, and encouraging students to take responsibility for their learning [26]. By providing clear academic expectations, timely feedback, and emotional support, teachers can help students manage their time more effectively and avoid delaying important academic tasks [27].

Further support for this idea comes from Patrick, Hisley, and Kempler [28], who demonstrated that enthusiastic teachers could significantly impact students' emotional and cognitive engagement in learning. This engagement, driven by positive teacher-student interactions, has been shown to reduce procrastination by encouraging students to stay on top of their academic responsibilities [29]. Enthusiasm and teacher motivation affect the educational aspect of learning and create an environment where students are more likely to self-regulate their emotions and focus on long-term goals rather than short-term distractions [16].

Laybourn, Frenzel & Fenzl [30] propose that teachers who create an autonomy-supportive classroom environment where students feel empowered and respected can reduce procrastination behaviors. When students perceive that their teachers trust and support their independence, they are likelier to engage in self-regulated learning. This, in turn, reduces procrastination tendencies because students feel a sense of ownership over their education [31]. They further argued, autonomy-supportive teaching has been linked to numerous positive

outcomes, including increased motivation, improved academic performance, and reduced procrastination.

On the contrary, demotivated or burned-out teachers can harm students' academic engagement. In their longitudinal study, Brouwers and Tomic, found that students of unmotivated teachers were more likely to procrastinate, disengage from their studies, and perform poorly academically. The result of their study indicated that low motivation among teachers often leads to a lack of enthusiasm and ineffective teaching methods, which can exacerbate procrastination among students. They suggest that motivated teachers inspire students, while demotivated teachers may unintentionally contribute to students' procrastination [32].

In a collectivistic society like Pakistan, family, and societal expectations play a significant role in shaping students' academic motivation. As Kaur and his colleague suggested, teachers and parents often impose family values or social norms on their children, emphasizing academic success to fulfill communal or familial responsibilities. This external pressure contributes to students' extrinsic regulatory motivation system, driving them to perform academically, not just for personal achievement but also to meet familial expectations. Similarly, teachers in this context reinforce societal expectations, making academic success a matter of family pride and social status, which can sometimes motivate students to avoid procrastination [33].

While different from intrinsic motivation, this kind of extrinsic motivation is essential in contexts where students may still need to embrace personal responsibility for their learning. Teachers who align their teaching with societal and familial expectations provide a crucial external influence. By guiding students to develop better study habits, regulate their emotions, and meet academic deadlines, teachers can help minimize. Teachers' motivation, therefore, plays a direct role in reducing procrastination, as it creates a structured, supportive learning environment where students are encouraged to take charge of their academic success. From the literature, we proposed

H1: Teachers' academic motivation has a significant and direct negative impact on students' procrastination behavior.

#### ***Mediating role of emotional regulation in TAM-SPB association***

In addition to motivation, a wide range of coping strategies are linked to academic procrastination. Emotion regulation and study habits are the most relevant strategies for producing desired learning outcomes [34]. Procrastination is often understood as a failure to regulate

emotional states, creating a gap between intention and action, which results in negative consequences for the individual. It is seen as the polar opposite of self-regulation and considered a deregulation of emotions that influences behavior, ultimately disrupting the academic process for university students [35].

Emotion regulation involves recognizing, understanding, and modifying one's emotional reactions to external stimuli, enabling individuals to maintain focus and balance [36]. Gross conceptualized it as, "the process by which individuals manage and adjust their emotional experiences and expressions to effectively meet their environment's demands". In another study Gross argued that effective emotion regulation requires the ability to understand and manage emotions and to employ strategies that help maintain emotional equilibrium, especially in stressful or challenging situations [37]. This ability plays a crucial role in preventing negative emotions such as anxiety, frustration, or boredom from interfering with academic focus and productivity.

Pychyl & Sirois in their study argued, poor emotional regulation elevates stress levels, impairing students' focus and productivity. Therefore, they concluded that high levels of procrastination have been associated with anxiety, stress, and negative impacts on both physical and mental health [38]. Students are more likely to delay academic tasks without proper emotion regulation, leading to detrimental academic outcomes. Pallini with his colleagues found that individuals often postpone or avoid tasks they perceive as unpleasant, anticipating that completing them might provoke aversive emotions like shame or guilt, leading to self-regulation breakdown and increased procrastination [39].

Additionally, procrastination may arise from negative emotional states and susceptibility to pleasurable temptations, as individuals prioritize short-term gratification over task completion [40]. Research has consistently shown that academic procrastination negatively affects students' education (e.g., academic performance: [41] health (e.g., psychological distress; Johansson et al., [42], and emotional well-being (e.g., negative emotions: Munda & Tiwari, [43]). In particular, emotional self-regulation has been shown to play a crucial role in reducing procrastination by enabling students to manage their feelings and reframe stressful situations into more manageable tasks.

Eckert et al. [44] explored the connection between procrastination and emotional distress, finding that poor emotion regulation contributes to procrastination by intensifying negative emotions. The study suggests that interventions to improve emotion regulation, such as mindfulness and cognitive reappraisal, can reduce

procrastination by alleviating the emotional distress associated with task avoidance. Similarly, Mohammadi Bytamar et al. [45] found that individuals who struggle with regulating unpleasant emotions, such as anxiety or irritation, are more likely to procrastinate, especially when they fear failure.

Interventions that focus on boosting emotional self-regulation have been shown to reduce procrastination. For instance, Quoidbach et al., [46] found that interventions designed to induce positive moods can enhance self-regulation capacity, enabling individuals to overcome procrastination. Likewise, self-affirmation interventions [47] can improve emotional resilience and self-regulation, reducing procrastination behaviors.

Understanding the mechanisms of emotion regulation is essential because effective emotion regulation helps students manage emotions such as anger, anxiety, and stress, preventing these feelings from impairing their decision-making abilities and increasing procrastination. Those with solid emotion regulation skills can better tolerate and modify negative emotions, allowing them to stay focused on completing academic tasks despite emotional challenges [48]. Based on the findings from previous research, it is proposed that emotion regulation mediates the relationship between teachers' academic motivation and student procrastination. Specifically, higher teacher motivation can help students develop better emotion regulation skills, reducing procrastination. From the above discussion, the following hypothesis was proposed:

H2: Emotional regulation mediates the relationship between teachers' academic motivation and student procrastination.

### ***Mediating role of effective study habits in TAM-SPB association***

Effective study habits are foundational to a student's academic success, profoundly shaping their overall performance [49]. These habits help students absorb and retain information and are critical mediating factors linking teachers' academic motivation with students' procrastination behavior. By bridging the gap between external motivation provided by teachers and the internal discipline required for academic success, study habits play a critical role in promoting academic achievement.

Nonis and Hudson [50] define study habits as strategies that students use to learn, such as paying attention in class, being punctual, taking comprehensive notes, completing assignments on time, and reviewing study material before lectures. These habits are likely to impact

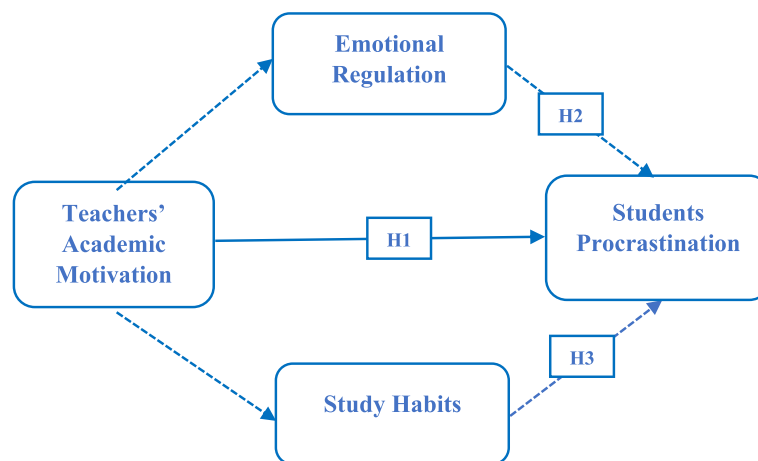
academic performance (p.230) significantly. Well-established habits, such as creating regular study schedules, engaging actively during lessons, practicing effective time management, and strategically reviewing materials, not only enhance comprehension but also shield students from negative academic consequences such as procrastination. These habits create a structured and organized approach to learning, allowing students to handle their academic responsibilities more efficiently and with less stress [51].

Garcia [52] and Crede and Kuncel [51] found a strong positive correlation between effective study habits and academic success. Similarly, Tus and his colleagues [53] pointed out that good study habits are often perceived as critical determinants of educational performance and serve to reduce procrastination among students. They further highlight that effective study habits equip students with the tools to approach tasks proactively, preventing the accumulation of overwhelming workloads that often lead to procrastination. On the contrary, poor study habits can lead to feeling overwhelmed, which fosters procrastination. Students who lack good study habits often need help with time management, resulting in stress, confusion, and missed opportunities to engage deeply with course materials [54]. Students who plan and organize their study schedules may stay caught up, perpetuating a cycle of avoidance and last-minute cramming.

Klassen et al. [21] explored the psychological mechanisms behind this connection, highlighting that student with strong self-regulation skills, often developed through good study habits, tend to manage their time more effectively and procrastinate less. This is supported by research showing that proactive study behaviors such as regularly reviewing material, setting short- and long-term academic goals, and self-assessment are critical factors in reducing procrastination and improving academic performance [55]. These findings illustrate the strong relationship between good study habits and students' ability to regulate their behavior and resist procrastination.

Numerous studies have underscored the importance of effective study habits in fostering academic success. Laakkonen & Nevgi [56] argued that students who plan their study schedules and adhere to them experience less stress and develop a deeper understanding of course materials. This sense of control over one's academic responsibilities promotes confidence and reduces the likelihood of procrastination.

Further supporting this, Hands and Limniou [57] also highlighted that study habits reflect a student's



**Fig 1** Conceptual framework

approach to learning, whether they adopt a deep learning approach characterized by engagement with the material and understanding its broader implications or a surface learning approach, typically leading to last-minute cramming. Deep learning is associated with better academic performance and reduced procrastination, as students who adopt this approach tend to engage more thoroughly with the material.

Faramida and colleagues [58], June investigated the correlation between study habits and academic achievement in senior secondary school students. The study found that students with good study habits consistently outperformed their peers with poor habits. This evidence reinforces that effective study habits are critical in fostering academic success and minimizing procrastination.

Kaur and Singh [59] researched female university students' study habits and concluded that many needed more effective study habits, which was detrimental to their academic performance. Based on the extensive literature that highlights the relationship between study habits, academic motivation, and procrastination behavior, effective study habits play a critical role in mitigating procrastination and enhancing academic success. Students who can manage their time well, engage actively with course materials, and establish consistent study routines are better equipped to succeed academically and less likely to procrastinate. Therefore, this study explores how study habits mediate the relationship between teachers' academic motivation and students' procrastination tendencies.

Given these insights, the following hypothesis is proposed:

H3: Effective study habits significantly mediate the relationship between teachers' academic motivation and students' procrastination behavior.

### Conceptual framework for current study

The (Figure1) presents a model examining how teachers' academic motivation influences students' procrastination behavior, both directly and indirectly. It hypothesizes that teachers' motivation directly affects students' procrastination (H1). Additionally, the model suggests that emotional regulation (H2) and study habits (H3) play significant mediating roles in this relationship. In essence, the proposed model highlighted that motivated teachers could reduce students' procrastination by directly influencing them and developing better emotional control and study practices (see Figure 1).

## Methodology

### Research design

The current study adopted a quantitative, cross-sectional survey design to investigate the impact of teachers' motivation on students' procrastination behavior, with students' study habits and emotion regulation serving as mediating variables. This design was chosen because it allowed for the efficient collection of data from a large sample at a single point in time, making it suitable for examining existing relationships between multiple variables.

### Population and sample

The study targeted all the permanent social science faculty members from three public universities in Multan. The population of faculty members across the three

**Table 1** Respondent’s profile

| University                                      | Number of Participants | Male (50%) | Female (50%) |
|-------------------------------------------------|------------------------|------------|--------------|
| Bahauddin Zakariya University (BZU)             | 130                    | 65         | 65           |
| Women University Multan                         | 35                     | 18         | 17           |
| Muhammad Nawaz Sharif University of Agriculture | 45                     | 23         | 22           |
| Total                                           | 210                    | 106        | 104          |

universities varied significantly: Bahauddin Zakariya University had 650 faculty members, Women University Multan had 175, and Muhammad Nawaz Sharif University of Agriculture had 230. To ensure that the sample from each university reflected these differences in size, the researchers divided the total population into strata (groups) based on the university each faculty member belonged to. The respondents were informed that their responses would be used solely for research purposes and assured that all information would remain confidential. To achieve a representative sample from each institution, a proportional stratified random sampling method was employed, ensuring that the size of the sample from each university reflected its share of the overall population. A 20% sample was taken from each university, resulting in 130 faculty members from Bahauddin Zakariya University, 35 from Women University Multan, and 45 from Muhammad Nawaz Sharif University of Agriculture. This sampling method guarantees an unbiased and proportionate representation across the institutions, with participants being randomly selected from each university. The data were collected during the fall semester from September 2023 to March 2024, using a structured questionnaire administered to university teachers in Multan, Pakistan. Moreover, the sample was stratified by gender to allow for detailed gender-based analysis, further enhancing the study’s comprehensiveness and objectivity. Additionally, the sample was stratified by gender to allow for detailed gender-based analysis, further strengthening the study’s comprehensiveness and objectivity (see Table 1).

Measures

A structured questionnaire was adapted and used to collect data from participants, which was divided into four key sections, each corresponding to the main variables of the study.

*Teachers’ Motivation Scale (Autonomous motivation for teaching)*

The "Autonomous Motivation for Teaching" scale was used for the present study to collect teacher data. The Autonomous Motivation for Teaching Scale is initially developed by Roth et al. [26]. This scale is grounded in

Self-Determination Theory (SDT). It is designed to assess the different types of motivation that drive teachers’ engagement in their profession and its potential impact on mitigating student procrastination. There are 16 items in total, and the scale is further divided into four primary subscales: Intrinsic Motivation, Identified Motivation, Introjected Motivation, and Extrinsic motivation. In the present study, we selected and adapted eight items, two from each subscale. Teachers responded on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree), allowing for the quantification of motivation levels. The adaptation process included expert validation and pilot testing to ensure relevance and clarity. The reliability of the adapted scale in this study was confirmed with a Cronbach’s alpha of ( $\alpha = 0.943$ ), demonstrating its suitability for the research context.

*Tuckman Procrastination Scale (TPS)*

The Tuckman Procrastination Scale (TPS) was used in this study to measure procrastination among university students, developed by Tuckman, [60]. It is a 16-item scale, with responses on a 5-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The maximum score is 64, and the minimum score is 16. Higher scores indicate a higher level of procrastination. A score below 20 signifies the absence of procrastination, placing the person among ‘non-procrastinators.’ Scores between 21 and 32 indicate mild procrastination, while scores between 33 and 47 reflect ample procrastination. Scores of 48 or higher denote severe procrastination, which could damage a person’s academic life [60]. The scale was adapted, and six statements were considered suitable for collecting data from university teachers regarding students’ procrastination behavior. The alpha reliability coefficient for this scale in the present study was ( $\alpha = 0.842$ ).

*Study Habits Inventory (SHI)*

The Study Habits Inventory (SHI), developed by Palsane and Sharma [61], was used as a mediating variable to examine the impact of students’ study habits on the relationship between teachers’ motivation and student procrastination. The adapted 4-item scale assesses critical areas such as time management, concentration, and

the study environment, using a 5-point Likert scale (1 = Never to 5 = Always). Adapted for this study, the scale underwent pilot testing to ensure relevance and clarity. The Cronbach's alpha for the scale was ( $\alpha = 0.799$ ), confirming its reliability for use as a mediator in this study.

### Emotion Regulation Questionnaire (ERQ)

The Emotion Regulation Questionnaire (ERQ), developed by Gross and John [62], was used as a mediating variable to assess the role of emotion regulation in the relationship between the independent and dependent variables. The 10-item scale measures two key emotion regulation strategies: cognitive reappraisal and expressive suppression. Participants responded on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree), capturing their emotional management tendencies. The scale was adapted with a 5-item final, validated for the study context, and its reliability, with a Cronbach's alpha of ( $\alpha = 0.835$ ), confirmed its suitability for use as a mediator in the analysis.

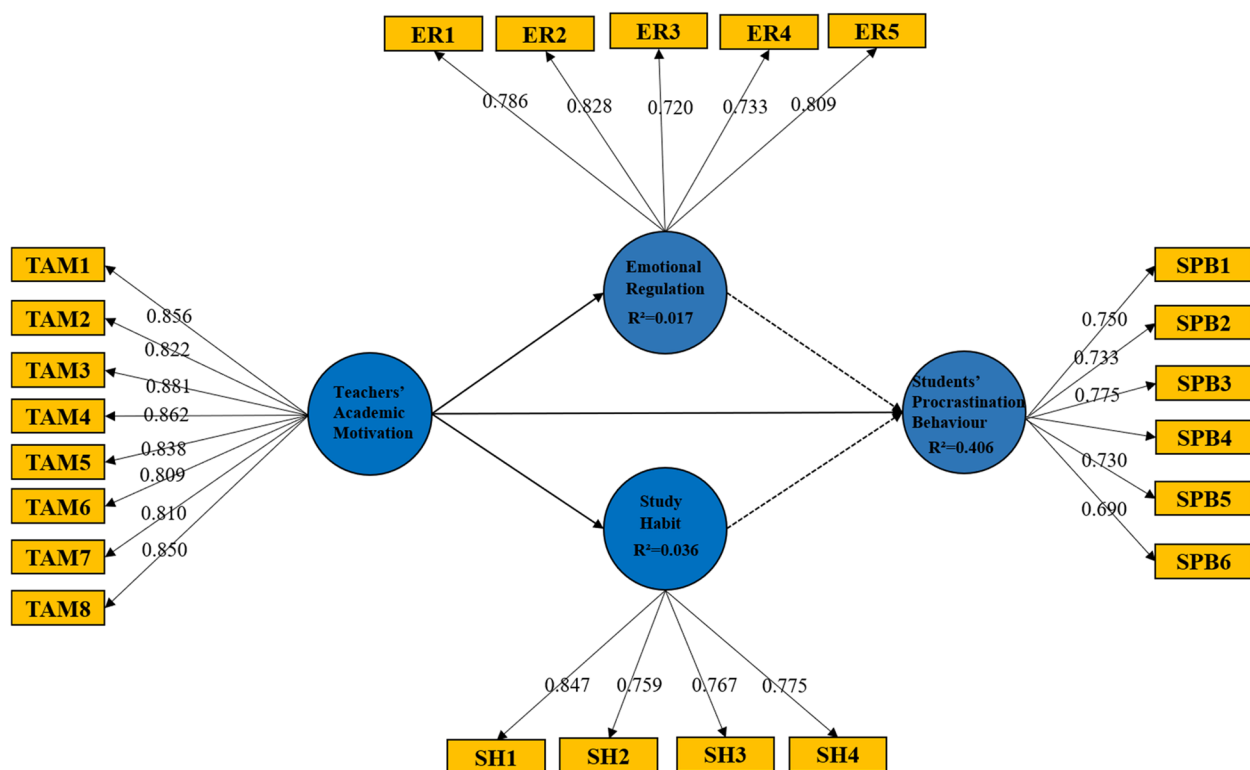
### Data analysis

According to previous research, Least Squares Structural Equation Modeling (PLS-SEM) is often used to explore the relationship between dependent and independent latent variables and is also considered one of

the most suitable tools for analyzing complex models [63]. Given the complexity of the proposed model for the current study, the (PLS-SEM) approach was used, employing Smart-PLS-4 software to test the proposed hypothesis. The SEM modeling procedure followed two key stages, as Hair et al. [64] outlined. The first stage focused on the psychometric properties of the measurement model (MM), ensuring the constructs used in the study (teachers' academic motivation, student procrastination behavior, emotion regulation, and study habits) met the necessary validity and reliability criteria. The second stage involved assessing the structural model (SM) to test the hypothesized relationships between the variables, particularly the mediating roles in the current study (emotion regulation and study habits mediating the relationship between teachers' academic motivation and student procrastination).

### Measurement Model (MM)

Both convergent and discriminant validity were assessed to ensure the quality of the measurement model, which commenced with the assessment of factor loading, as shown in (Figure 2). For convergent validity, Factor Loadings (FL), Average Variance Extracts (AVEs), and Composite Reliability (CR) were evaluated, and Cronbach's Alpha was also examined (Tables 2 and 3). Hair et al. [63]



**Fig 2** Measurement model

**Table 2** Factor loading (FL) and Variance inflation factor (VIF)

| Constructs                        | Item Coding | TAM   | SPB   | ER    | SH    | VIF   |
|-----------------------------------|-------------|-------|-------|-------|-------|-------|
| Teachers Academic Motivation      | TAM1        | 0.856 |       |       |       | 3.118 |
|                                   | TAM2        | 0.822 |       |       |       | 3.883 |
|                                   | TAM3        | 0.881 |       |       |       | 3.472 |
|                                   | TAM4        | 0.862 |       |       |       | 3.158 |
|                                   | TAM5        | 0.838 |       |       |       | 3.712 |
|                                   | TAM6        | 0.809 |       |       |       | 3.970 |
|                                   | TAM7        | 0.810 |       |       |       | 4.152 |
|                                   | TAM8        | 0.850 |       |       |       | 4.227 |
| Students Procrastination Behavior | SPB1        |       | 0.750 |       |       | 1.753 |
|                                   | SPB2        |       | 0.733 |       |       | 1.613 |
|                                   | SPB3        |       | 0.775 |       |       | 1.759 |
|                                   | SPB4        |       | 0.808 |       |       | 1.979 |
|                                   | SPB5        |       | 0.730 |       |       | 1.541 |
|                                   | SPB6        |       | 0.690 |       |       | 1.443 |
| Emotion Regulation                | ER1         |       |       | 0.786 |       | 1.583 |
|                                   | ER2         |       |       | 0.828 |       | 4.376 |
|                                   | ER3         |       |       | 0.720 |       | 1.511 |
|                                   | ER4         |       |       | 0.733 |       | 1.524 |
|                                   | ER5         |       |       | 0.809 |       | 4.126 |
| Study Habits                      | SH1         |       |       |       | 0.847 | 2.000 |
|                                   | SH2         |       |       |       | 0.759 | 2.674 |
|                                   | SH3         |       |       |       | 0.767 | 1.738 |
|                                   | SH4         |       |       |       | 0.775 | 2.860 |

Note: TAM Teachers' Academic Motivation, SPB Students Procrastination Behavior, EM Emotion Regulation, ST Study Habit

**Table 3** Construct validity and reliability

| Constructs | Cronbach's alpha | rho_a | rho_c | AVE   |
|------------|------------------|-------|-------|-------|
| TAM        | 0.943            | 0.963 | 0.951 | 0.708 |
| ER         | 0.835            | 0.843 | 0.883 | 0.20  |
| SH         | 0.799            | 0.815 | 0.867 | 0.560 |
| SPB        | 0.842            | 0.845 | 0.884 | 0.708 |

Note: AVE Average Variance Extraction, rho\_a, rho\_c Composite Reliability

**Table 4** HTMT analysis

|     | TAM   | ER    | SH    | SBP |
|-----|-------|-------|-------|-----|
| ER  | -     | -     | -     | -   |
| SH  | 0.245 | -     | -     | -   |
| SBP | 0.651 | 0.483 | -     | -   |
| TAM | 0.136 | 0.207 | 0.233 | -   |

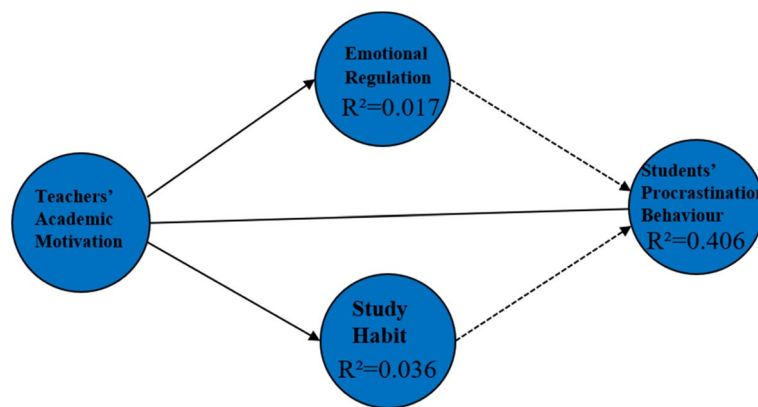
Note: HTMT Heterotrait-Monotrait ratio

state that factor loading and composite reliability values above 0.70 and AVE values above 0.5 are considered acceptable. For discriminant validity,

Heterotrait-Monotrait (HTMT) technique was employed (Table 4). As per Henseler et al. [65], HTMT values below 0.90 are considered acceptable to confirm the discriminant validity of the construct.

In Table 2, a convergent validity evaluation revealed that all factor loadings (FL) exceeded the threshold of 0.7, indicating solid associations with their respective constructs and each item reliably measuring its intended construct [63]. Additionally, the Collinearity test is applied to see all indicators' variance inflation factor (VIF). The VIF values below five are considered safe for ensuring good validity of constructs [66]. VIF values, ranging from 1.443 to 4.376, suggest moderate multicollinearity for some items but remain within acceptable limits, indicating that the items are appropriate for measuring the constructs and that the overall model is robust.

In Table 3, the Average Variance Extracted (AVE) for each construct surpassed the 0.5 mark, demonstrating substantial variance attributable to the constructs themselves, indicative of good construct validity [67]. The Composite Reliability (CR) values, exceeding the 0.7 threshold, and Cronbach's Alpha (CA) values



**Fig 3** Structural model

further affirmed the internal consistency and reliability of the scales [68].

Table 4 demonstrated the Heterotrait-Monotrait Ratio (HTMT) values, which were below the conservative thresholds of HTMT .85 and HTMT .90 [65]. This differentiation ensures that the constructs are distinct and measure different dimensions effectively. This differentiation is essential to ensure that the constructs are distinct, which could confound the analysis of their interrelationships.

### Structural Model (SM)

The structural model in the PLS-SEM testing hypothesis with the PLS techniques is always estimated by looking at the path coefficient, *t* statistic, standard error, and the amount of  $R^2$ , which can explain the strength and direction of the relationship and significance. Meanwhile, the *t*-statistic and standard error are used to analyze the magnitude of the effect [63]. The value of  $R^2$  indicates the amount of variance explained. The variances linked with the dependent variables determined the suggested model's explanatory power. The Bootstrapping procedure was used to analyze the robustness of the factor loadings and the significance between variables, as shown in (Figure 3). This study uses the 5000 bootstrap resampling technique [64] to determine whether the hypothesis is accepted. The hypothesis would be supported at the 0.05 significance level if the *t*-statistic exceeds the critical value.

### Hypothesis testing

Hypothesis testing is the final stage in the structural model to test the relationship between constructs (teachers' academic motivation, students' procrastination behavior, emotion regulation, and study habits). This study consisted of three initial hypotheses (see Table 5). All the hypotheses are supported by an analysis using the

structural model. A detailed explanation is carried out below;

H1: Teachers' Academic Motivation is significantly positively correlated with Students Procrastination Behavior

The results for H1 revealed a strong positive and significant relationship between teachers' academic motivation to reduce students' procrastination behavior, with a beta coefficient of 0.105, indicating a substantial impact. This relationship was supported by a *t*-value of 3.056 and a *p*-value of 0.002, demonstrating high statistical significance. This finding aligns with prior research, such as Serdar et al., [69], which emphasized that empathetic teacher-student interactions significantly enhance student learning outcomes by creating a supportive and responsive environment.

H2: Teachers' Academic Motivation and Students' Procrastination Behavior Mediated with Emotion Regulation

In testing the mediated effect of emotion regulation on the relationship between teachers' academic motivation and students' procrastination behavior, H2 indicated a positive mediated pathway, albeit weaker than the direct effects. The beta was 0.063, with a *t*-value of 2.037 and a *p*-value of 0.042, pointing to statistical significance but lower impact strength. This outcome supports research like Rezaei& Zebardast [70] that emphasizes the importance of students feeling valued and accepted in enhancing educational outcomes.

H3: Teachers' Academic Motivation and Students' Procrastination Behavior Mediated with Study Habits

**Table 5** Hypothesis testing results

|    | Relationships    | $\beta$ | M     | SD    | t-value | p-value | Decision  |
|----|------------------|---------|-------|-------|---------|---------|-----------|
| H1 | TAM -> SPB       | 0.105   | 0.106 | 0.034 | 3.056   | 0.002   | Supported |
| H2 | TAM -> ER -> SPB | 0.063   | 0.066 | 0.031 | 2.037   | 0.042   | Supported |
| H3 | TAM -> SH -> SPB | 0.053   | 0.055 | 0.021 | 2.567   | 0.010   | Supported |

In testing the mediated effect of study habits on the relationship between teachers' academic motivation and students' procrastination behavior, H3 indicated a positive mediated pathway, albeit weaker than the direct effects. The beta was 0.053, with a t-value of 2.567 and a p-value of 0.010, pointing to statistical significance but lower impact strength. This outcome supports research like Hadadan& Ghajareih [71]. that emphasizes the importance of students feeling valued and accepted in enhancing educational outcomes.

## Discussion and implications

Academic procrastination is one of the essential factors in education sustainability. Psychologists and educational researchers found procrastination to be a potentially harmful behavior among students, and they always remained busy finding ways to overcome its potential effects. Initially, studies on procrastination behavior focused on psychological factors such as anxiety and stress and social factors like family pressure and inadequate study environments. However, little attention was given to the role of teachers as a source of external motivation for students. The present study aimed to analyze the combined external and personal regulatory potential to reduce students' procrastination behavior, with emotion regulation and study habits as mediating factors in previous studies [72, 73]. This research investigated the influence of teachers' academic motivation, emotion regulation, and regulatory study habits on student procrastination behavior. Based on the findings obtained, to highlight the primary results, it has been corroborated that teachers' academic motivation ( $H^1$ ) has a significant negative relationship with procrastination behavior. One possible explanation for this outcome is that students are more motivated to engage in learning when they have meaningful interactions with teachers and peers. In the presence of their classmates, students are inclined to improve their skills and maximize learning opportunities, which reduces their tendency to procrastinate.

Additionally, teachers play a crucial role in fostering students' potential, especially when they feel unmotivated [74]. Furthermore, the study also supported the second and third hypotheses. It indicated that when students feel a sense of personal responsibility for completing a task, they tend to regulate their emotions more

effectively and adopt better study habits, such as time management, goal setting, and prioritizing tasks, which help reduce their tendency to procrastinate.

## Theoretical contributions

This research provides a valuable contribution to the theoretical understanding of procrastination by integrating external and internal factors that influence student behavior. By incorporating teachers' academic motivation as an external motivator and exploring the mediating roles of emotion regulation and study habits, this study expands the theoretical landscape beyond the individual-focused models of procrastination. The findings also support that teachers play a pivotal role in shaping the academic environment, offering further insight into how teacher motivation can influence student self-regulation. These contributions align with the Theory of Planned Behavior (TPB) which emphasizes the importance of external motivational factors in promoting student engagement and reducing procrastination.

## Practical implications

As a result of this finding, educational strategies should prioritize enhancing teacher training programs that foster motivation and student engagement rather than merely concentrating on structural or administrative improvements. By cultivating these qualities in teachers, students are more likely to feel supported and motivated, leading to better academic outcomes. Institutions should consider designing workshops and training sessions that empower teachers to enhance students' emotional regulation and study habits, encouraging a more proactive approach to learning. The findings suggest that educational policies should focus on academic content and on implementing policies that support personalized learning plans, collaborative projects, and peer interaction to reduce procrastination tendencies.

## Limitations and recommendations of the study

One limitation of the present study is the relatively limited sample size, which may affect the generalizability of the findings across different educational contexts. Future research should consider more extensive and diverse

samples, including students from various universities, disciplines, and regions, to validate the results in broader contexts.

The study used a cross-sectional research design, which limits the ability to conclude causal relationships over time. While the findings indicate significant associations, a longitudinal design would provide deeper insights into how teachers' academic motivation and students' self-regulation evolve and their long-term effects on procrastination behavior.

While the present study focused on emotion regulation and study habits as mediators between TAM and SPB, future studies could explore additional mediators, such as self-efficacy, academic resilience, or goal orientation, to understand more comprehensively how teacher motivation influences student outcomes.

### Future direction

Future research could address the limitations of this study by incorporating longitudinal designs to explore the long-term effects of teacher academic motivation and students' self-regulation on procrastination. It would also be valuable to examine additional mediators, such as self-efficacy, academic resilience, or goal orientation, to understand further how teacher motivation influences student outcomes. Moreover, future studies could investigate the effectiveness of interventions to improve teacher-student interactions, emotional regulation, and study habits. Implementing and testing specific strategies for improving teacher motivation and student engagement would offer valuable insights into practical approaches for reducing procrastination in academic settings.

### Conclusion

The present study has demonstrated the significant role that teachers' academic motivation plays in reducing students' procrastination behavior, with emotion regulation and study habits acting as critical mediators. The findings emphasize the importance of fostering positive teacher-student interactions, where teachers serve as external motivators, encouraging students to engage in their academic responsibilities more effectively. By helping students regulate their emotions and develop strong study habits, teachers can play a pivotal role in combating procrastination and promoting academic success. These insights call for re-evaluating teacher training programs, encouraging a shift towards strategies that enhance student engagement and motivation. Furthermore, addressing personal and external regulatory factors is critical to creating a sustainable educational environment where

procrastination is minimized, and students are empowered to reach their full potential.

### Supplementary Information

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Supplementary Material 1.

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### Informed consent

Informed consent was obtained from all individual adult participants included in the study.

### Authors' contributions

**Authors Contribution** All authors have read and agreed to the published version of the manuscript. Sidra Sarwar wrote the initial draft, conceptualized the model, performed Formal analysis, curated Data, and helped with data interpretation. Anam Nazneen Tara helped in the revise draft, analysis section, and editing the draft. Muhammad Naseem Abid administered the data analysis, data supervision, interpretation, and Visualization. Suad Dukhaykh helped in reviewing the revise draft, proof reading, formatting, interpretation and supervising.

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### Data availability

For inquiries concerning the data and materials presented in this paper, please feel free to contact the corresponding authors via email at [anamtara@hust.edu.cn](mailto:anamtara@hust.edu.cn); [abid@hust.edu.cn](mailto:abid@hust.edu.cn)

### Declarations

#### Ethics approval and consent to participate

"Teachers' Academic Motivation and Student Procrastination Behavior: Mediating Effects of Emotion Regulation and Study Habits". This research was approved by the Research Committee (No: Soe/Hust-April-2024-66) of the School of Education, Huazhong University of Science and Technology, Wuhan, Hubei, China, in accordance with the inter-national ethical standards, Helsinki Declaration 1964 and its amendments.

#### Consent for publication

Not applicable.

#### Competing interests

The authors declare no competing interests.

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